Alteryx Workflow Document

Problem_0003.yxmd

Report Create Date : 2026-04-25

Workflow Basic Information

FileName : Problem_0003.yxmd
Workflow name : Problem_0003
yxmd version : 2025.1
RunE2 : T
Workflow description :

Author Information

Author :
Company :
Copyright :

Workflow Meta information

Setting Name	Value
Root Tool Name	
Tool Version	
Tool In DB	
Tool In DB value	False
Category Name	
Search Tags	
Description Link actual	
Description Link displayed	
Workflow ID	6d1aa9fc-55e9-4423-b0e7-575343e4b820
Platform Workflow ID	
Example Description	
Example File	
Telemetry Previous Workflow ID	dc9f9483-ac94-49e0-852c-48e6588efdc9
Telemetry Origin Workflow ID	dc9f9483-ac94-49e0-852c-48e6588efdc9

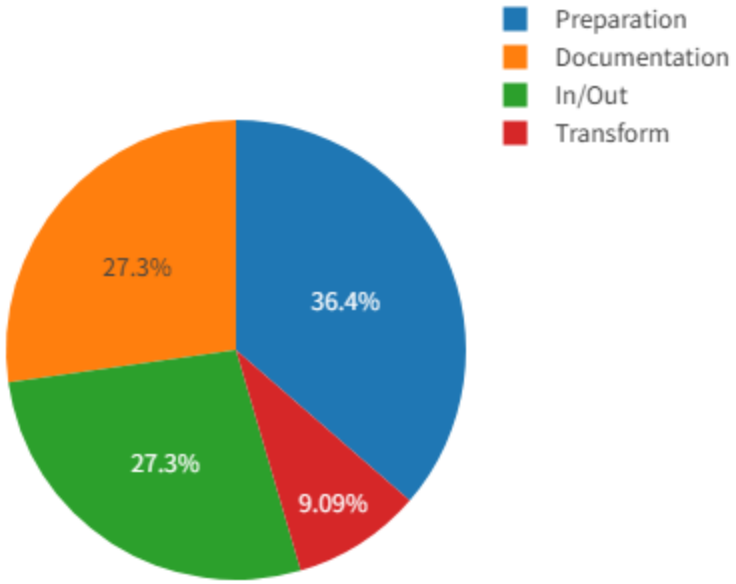
workflow detail setting

Setting Name	Value
yxmdVer	2025.1
RunE2(AMP Engine)	T
Memory_default	True
Global Record Limit	0
Temp Files	True
Annotation on	True
Annotation include Tool Name	False
Convert Error Limit	10
Convert Error Limit Stop	False
Cancel On Error	False
Disable Browse tool	False
Enable Performance Profiling	False
Run With E2	True
Sorted Grouping(Engine Compatibility Mode)	True
Default TZ	Local
Predictive Tools Code Page	932
Disable All Output	False
Show All Macro Messages	False
Show Connection Status Is On	True
Show Connection Status Only When Runninng	True
Zoom Level	0
Layout Type	Horizontal
Is Template	False
Name Is File Name	True

Tool statistics

Total Tool Count : 11
Number of tool types : 6

Category



Tool Category	Count
Preparation	4
Documentation	3
In/Out	3
Transform	1

Tools

Tool Category	Tool Name	Count
Documentation	Comment	3
In/Out	Text Input	2
Preparation	Filter	2
Preparation	Generate Rows	2
In/Out	Browse	1
Transform	Summarize	1

Input/Output

Input overview

Tool Name	ToolID	Input Type	Input
Text Input	2	Text	Text
Text Input	4	Text	Text

Output overview

Tool Name	ToolID	Output Type	Output
Browse	10	Browse	

Workflow Canvas Image

Alteryx Workflow detail description

Workflow Overview

This Alteryx workflow is designed to find the largest prime factor of a given large integer. It accomplishes this by first identifying all potential divisors up to the square root of the number, then filtering these to find actual divisors. Subsequently, it performs a primality test on each actual divisor to determine if it is a prime number. Finally, it extracts the maximum value among these prime divisors.

Input/Output Data Overview

The workflow's input is a single hardcoded integer, 600851475143, provided by a Text Input tool. The output is a single value representing the largest prime factor of this input number, which is displayed in a Browse tool.

Workflow Details

Section 1: Data Initialization

This section introduces the target number into the workflow and generates a series of odd numbers that will serve as potential divisors for the target number.

Tool Name	Tool ID	Processing Details
Text Input	4	Injects the large integer 600851475143 into the workflow as a field named 'Number'.
Generate Rows	1	Generates a sequence of odd numbers, starting from 1 and incrementing by 2, up to the square root of the 'Number' field. These generated numbers are assigned to a new field called 'RowCount' and represent potential factors.

Section 2: Finding Divisors

This section filters the generated potential factors to identify which ones are actual divisors of the original input number.

Tool Name	Tool ID	Processing Details
Filter	3	Evaluates if the 'Number' (original input) is perfectly divisible by 'RowCount' (potential factor). Only records where 'Number' modulo 'RowCount' equals 0 are passed to the 'True' output, indicating 'RowCount' is an actual divisor.

Section 3: Primality Testing

For each actual divisor found, this section conducts a test to determine if that divisor is a prime number.

Tool Name	Tool ID	Processing Details
Generate Rows	5	For each 'RowCount' (divisor) received, this tool generates a new sequence of numbers, 'RowCount2', starting from 2. It continues to increment 'RowCount2' by 1 as long as 'RowCount' is not divisible by 'RowCount2' and 'RowCount' is greater than or equal to 'RowCount2'. This loop identifies if a 'RowCount' has any divisors other than 1 and itself.

Tool Name	Tool ID	Processing Details
Filter	6	Filters the output of the previous 'Generate Rows' tool. It passes records where 'RowCount' is equal to 'RowCount2' + 1. This condition is met when 'RowCount2' has iterated up to 'RowCount - 1' without finding a divisor, thus indicating that 'RowCount' is a prime number (assuming 'RowCount' > 2).

Section 4: Aggregation and Output

This final section determines the largest prime factor among those identified and displays it as the workflow's result.

Tool Name	Tool ID	Processing Details
Summarize	14	Takes all the prime factors ('RowCount') identified by the preceding steps and calculates the maximum value among them. The resulting field is renamed to 'Max_RowCount'.
Browse	10	Displays the final output, which is the single record containing the maximum prime factor identified by the workflow.